Question # 7 of 10 (Start time: 02:19:09 PM, 20 August 2021)

Ideally, the width of confidence interval should be

Select the correct option

- ○ 0
- ● 1
- ○ 100
- ○ 98

Question # 1 of 10 (Start time: 10:19:39 PM, 20 August 2021) Total Marks: 1

_____ is a range of numbers inferred from the sample that has a certain probability of including the population parameter over the long run.

Select the correct option

- Hypothesis
- Lower limit
- Confidence interval
- Probability limit

Question # 3 of 10 (Start time: 01:39:42 PM, 21 August 2021) Total Marks: 1

A confidence interval has a specified probability_____ of containing the true value of the parameter.

ect the correct option

- α
- α/2
- 1+α
- 1-α

Question # 3 of 10 (Start time: 01:27:41 PM, 21 August 2021)

Total Marks: 1

The Maximum Likelihood Estimators (MLE) are ..... and ..... but not necessarily .....

Select the correct option

- Consistent, efficient, unbiased
- Consistent, unbiased, efficient
- Unbiased, efficient, consistent
- Unbiased, consistent, efficient

An estimator is said to be efficient if it has

Select the correct option

○	Smallest Variance
○	Unbiased estimate
●	Both (a) & (b)
○	None of these

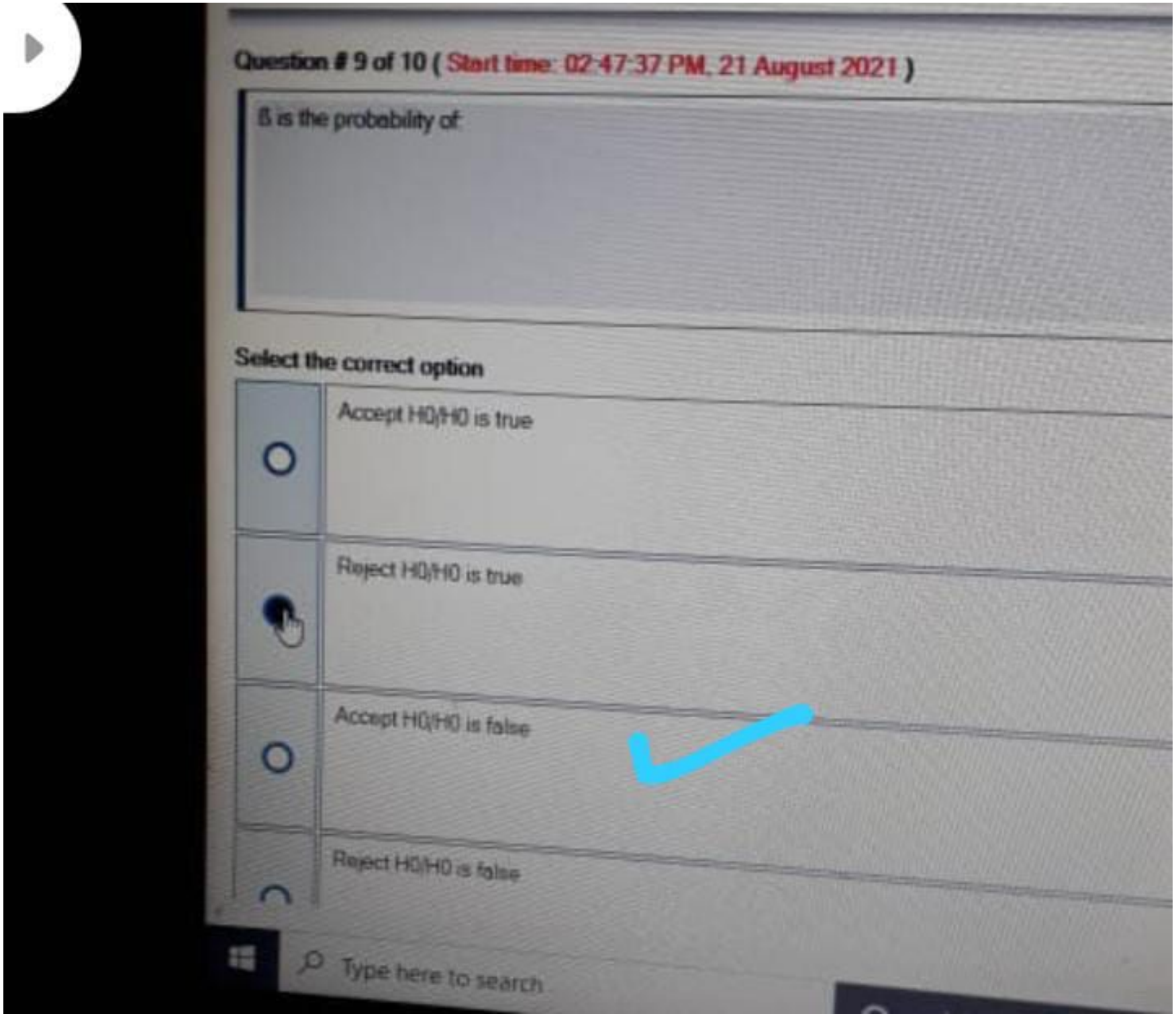
A statistic whose standard error decreases with an increase in the sample size will be .....

Select the correct option

- Efficient
- None
- Consistant
- Unbiased

beta is the probability of:

accepting H(not) actually H(1) is true


Question # 9 of 10 (Start time: 02:47:37 PM, 21 August 2021)
ß is the probability of
Select the correct option
Accept H0|H0 is true
Reject H0|H0 is true
Accept H0|H0 is false
Reject H0|H0 is false
Type here to search

STA301:Quiz 1 Quiz Start Time: 10:34 AM, 18 August 20

Question # 1 of 10 (Start time: 10:34:23 AM, 18 August 2021) Total Mark

The width of the confidence interval is called_____ of the estimate.

Select the correct option

- (●) Unbiasedness
- ( ) Precision
- ( ) Accuracy
- ( ) Biased

Question # 5 of 10 (Start time: 01:31:17 PM, 21 August 2021) Total Marks: 1

A confidence interval has a specified probability_____ of containing the true value of the parameter.

Select the correct option

- 1-α
- α/2
- α
- 1+α

Question # 10 of 10 (Start time: 01:37:24 PM, 21 August 2021) Total Marks: 1

"A failed student was passed by the examiner", is an example of:

Select the correct option

- Type-I error
- Type-IV error
- Type-III error
- Type-II error

Question # 3 of 10 (Start time: 01:49:07 PM, 21 August 2021)

Inferential statistics involves:

Select the correct option

- Testing of hypothesis
- All of these
- Confidence interval
- Estimation

Question # 7 of 10 (Start time: 01:44:56 PM, 21 August 2021) Total Marks: 1

_____ are equivalent.

he correct option

- Type-II error and Level of confidence
- Type-I error and Level of significance ✓
- Type-II error and Level of significance
- Type-I error and Level of confidence

Question # 7 of 10 (Start time: 01:33:27 PM, 21 August 2021)

Total Marks: 1

In a Geometric distribution, Maximum Likelihood Estimator (MLE) for proportion (P) is equal to:

Select the correct option

- Sample variance
- Sample mean
- Reciprocal of the sample variance
- Reciprocal of the mean ✓

Question # 5 of 10 (Start time: 07:42:21 PM, 21 August 2021) Total Marks: 1

How can we interpret the 90% confidence interval for the mean of the normal population.

Select the correct option

- There are 100% chances of falling true value of the parameter
- There are 90% chances of falling true value of the parameter
- All are correct
- There are 10% chances of falling true value of the parameter

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 01:42:17 PM, 21 August 2021) Total Marks: 1

From which of the following methods we can obtain point estimate of the population parameters.

e correct option

- All of the above
- Maximum Liklihood Method
- Method of Moments
- Method of Least Squares

Question # 4 of 10 (Start time: 04:19:24 PM, 21 August 2021)

The end points that bound the confidence interval are called

Select the correct option

- Lower limits
- Bounded limits
- Lower and upper limits
- Upper limits

Question # 3 of 10 (Start time: 04:18:34 PM, 21 August 2021)

Conventionally, the probability of making a type-I error is denoted by which of the following symbol?

Select the correct option

- s (sigma)
- ß (beta)
- ? (theta)
- a (alpha)



ß is the probability of:

Select the correct option

- Reject H0/H0 is true
- Reject H0/H0 is false
- Accept H0/H0 is false
- Accept H0/H0 is true

Question # 2 of 10 (Start time: 04:17:48 PM, 21 August 2021)

An estimator is said to be _____ if its expected value is equal to true value of its parameter.

Select the correct option

- ○ Efficient
- ○ Consistant
- ○ Biased
- ○ Unbiased

Question # 8 of 10 (Start time: 10:44:45 AM, 18 August 2021) **Total Marks: 1**

The method of Maximum Likelihood (ML) is used to find out:

Select the correct option

○	Sample values
○	Random numbers
○	Correlation coefficient
●	Point estimates

"A point estimate plus/minus a few times the standard error of that estimate". This statement represents:

Select the correct option

- ○ Acceptance region
- ○ Critical region
- ● Confidence interval
- ○ Critical value

Question # 3 of 10 (Start time: 10:37:27 AM, 18 August 2021) **Total Marks: 1**

The larger the n, the smaller the standard error, and so the narrower the

Select the correct option

○	None of the above
●	Confidence level
○	Level of significance
○	Confidence Interval

Question # 3 of 10 (Start time: 09:47:32 PM, 21 August 2021) Total Marks: 1

"To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as

Select the correct option

- (●) The method of least square
- ( ) The method of fractional moments
- ( ) The method of moments
- ( ) The method of maximum likelihood



STA301:Quiz 1

Question # 6 of 10 (Start time: 12:39:31 AM, 18 August 2021)

In Binomial distribution the sample size is considered to be sufficiently large, if both np and nq are greater than or equal to:

Select the correct option

- ○ 10
- ○ 3
- ● 5
- ○ 15

Question # 10 of 10 (Start time: 10:47:05 AM, 18 August 2021) **Total Marks: 1**

Which of the following is most important and most widely used method in point estimation?

Select the correct option

○	The method of moments
○	The method of fractional moments
○	The method of maximum likelihood
●	The method of least square

Question # 9 of 10 (Start time: 10:45:33 AM, 18 August 2021) **Total Marks: 1**

The precision of an estimator can be increased by increasing the:

Select the correct option

- ○ Size of sample
- ○ Number of estimators
- ● Size of population
- ○ Number of parameters

$n = 16$, $\bar{x} = 19.92$, $\sigma = 5.73$

$1 - \alpha = 95\% = 0.95$

$\alpha = 0.05 \Rightarrow \alpha/2 = 0.025$

we know that..

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$$19.92 \pm z_{0.025} \frac{5.73}{\sqrt{16}}$$

$$19.92 \pm 1.96 \left(\frac{5.73}{4}\right)$$

$$19.92 \pm 2.8077$$

$19.92 - 2.8077$, $19.92 + 2.8077$

17.1123, 22.7277

$(17.1123, 22.7277)$

Question # 10 of 10 (Start time: 10:28:59 PM, 20 August 2021) Total Marks: 1

The precision of an estimator can be increased by increasing the;

Select the correct option

- Level of significance
- Number of estimators
- Number of parameters
- Size of sample

Question # 5 of 10 (Start time: 05:16:26 PM, 22 August 2021) Total Marks: 1

If Var(mean1) < Var(mean2) it implies:

Select the correct option

- Mean1 is more efficient
- Mean2 is more efficient
- Mean1 is more consistent
- Mean2 is more consistent

Question # 4 of 10 (Start time: 05:15:23 PM, 22 August 2021) Total Marks: 1

Method of Moments for estimation of parameters is based on the theory of

Select the correct option

- Linear Estimation
- Non-linear estimation
- Both Linear and Non-linear
- None of the above

STA301:Quiz 1

Quiz Start Time: 11:59 AM, 22 August 2021

Question # 5 of 10 (Start time: 12:04:16 PM, 22 August 2021)

Total Marks: 1

If we have the following information,
P= 0.16, q = 0.84, Z=1.96 and e = 0.05, then sample size will be:

e correct option

- 200
- 205
- 199
- 207

Question # 4 of 10 (Start time: 01:28:18 PM, 21 August 2021)

Total Marks: 1

As sample size goes up, what tends to happen to 95% confidence intervals?

Select the correct option

- They become more narrow
- They become more precise and narrow
- They become more precise
- They become wider

Question # 1 of 10 (Start time: 05:12:15 PM, 22 August 2021) Total Marks: 1

The end points that bound the confidence interval are called;

Select the correct option

- Lower and upper limits
- Bounded limits
- Upper limits
- Lower limits

Question # 4 of 10 (Start time: 01:40:28 PM, 21 August 2021) Total Marks: 1

When sample size is to be considered large, population standard deviation can be replaced by

he correct option

- Sample Variance
- Sample Mean
- Population variance
- Sample Standard Deviation

Quiz 1

Quiz Start Time: 11:59 AM, 22 August 2021

n # 3 of 10 (Start time: 12:02:54 PM, 22 August 2021)

Total Marks: 1

If an estimator is more efficient then the other estimator, its shape of the sampling distribution will be

Select the correct option

- Flattered
- Normal
- Skewed to right
- Highly peaked

STA301:Quiz 1

Quiz Start Time: 04:31 PM, 22 August 2021

Question # 6 of 10 (Start time: 04:40:03 PM, 22 August 2021)

Total Marks: 1

A statistic whose standard error decreases with an increase in the sample size will be .....

the correct option

- None
- Unbiased
- Consistant
- Efficient

Question # 2 of 10 (Start time: 01:48:09 PM, 21 August 2021) Total Marks

The larger the standard error, the ..... the confidence interval.

correct option

- Narrower
- Wider
- Accurate
- Inaccurate

Question # 2 of 10 (Start time: 05:12:59 PM, 22 August 2021) Total Marks: 1

Interval estimation and Confidence interval are :

Select the correct option

- Different
- Dependent
- Same
- Independent



Question # 1 of 10 (Start time: 01:37:44 PM, 21 August 2021)

Total Marks: 1

Interval estimation and Confidence interval are :

e correct option

- Different
- Independent
- Dependent
- Same

Question # 6 of 10 (Start time: 06:31:53 AM, 22 August 2021) Total Marks: 1

We can apply method of Maximum Likelihood on:

Select the correct option

- Qualitative variables only
- Discrete as well as continuous variables
- Continuous variables only
- Discrete variables only

...A301:Quiz 1

Question # 4 of 10 (Start time: 12:38:14 AM, 18 August 2021)

If Var(mean1) < Var(mean2) it implies:

Select the correct option

- ○ Mean1 is more consistent
- ○ Mean2 is more consistent
- ○ Mean2 is more efficient
- ○ Mean1 is more efficient ✓

BC190201393: QURAT-UL-AIN SADIQ

STA301 Quiz 1

Question # 10 of 10 (Start time: 01:20:11 PM, 22 August 2021)

Quiz Start Time: 01:19 PM, 22 August 2

"To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as

Total Marks

Select the correct option

- [x] The method of maximum likelihood
- [ ] The method of least square
- [ ] The method of moments
- [ ] The method of fractional moments

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quiz.vu.edu.pk/QuizQuestion.aspx?ver=8aedac59-e71a-4898-9f99-afe489bcf7b0

Question # 7 of 10 (Start time: 09:54:44 PM, 21 August 2021)

Which of the following is not a criterion for a good problem situation?

Select the correct option

- It must be such that its solution will be benefited by group effort.
- It must be appropriate to the students' level of development.
- It must be authentic.
- It must be clearly structured and defined.

[Music]

Type here to search

Question # 10 of 10 (Start time: 01:22:28 PM, 22 August 2021) Total Marks: 1

In a Poisson distribution, Maximum Likelihood Estimator (MLE) for meu (u) is equal to:

Select the correct option

- ○ Reciprocal of the sample variance
- ○ Sample mean
- ○ Sample variance
- ● Reciprocal of the mean

STA301:Quiz 1

Question # 3 of 10 (Start time: 02:09:57 PM, 20 August 2021)

As sample size goes up, what tends to happen to 95% confidence intervals?

Select the correct option

- They become more narrow
- They become wider
- They become more precise and narrow
- They become more precise

Question # 2 of 10 (Start time: 06:27:56 AM, 22 August 2021) Total Marks: 1

In interval estimation we obtained a .....of values as an estimate of parameter.

Select the correct option

- Both a and b
- Range
- Group
- None of these

BC190201393: QURAT-UL-AIN SADIQ

STA301 Quiz 1

Question # 2 of 10 (Start time: 01:11:45 PM, 22 August 2021)

Which of the following relationship exist between sampling distribution of samples mean and sampling distribution samples median?

Select the correct option

- Var(median)< Var(median)
- None of these
- Both of these
- Var(mean)<Var(median)



STA301:Quiz 1

Question # 1 of 10 (Start time: 02:54:02 PM, 21 August 2021)

In the method of moments, how many equations are required for finding two unknow

Select the correct option

- 2
- 4
- 1
- 3

Question # 3 of 10 (Start time: 05:39:18 PM, 22 August 2021)

Alpha is the probability of

Select the correct option

- All of these
- Type I error
- Reject H0/H0 is true
- Committing Type I error

STA301:Quiz 1

Question # 9 of 10 (Start time: 12:43:09 AM, 18 August 2021)

How can we interpret the 90% confidence interval for the mean of the normal population.

Select the correct option

- ○ There are 10% chances of falling true value of the parameter
- ○ All are correct
- ○ There are 90% chances of falling true value of the parameter
- ○ There are 100% chances of falling true value of the parameter



Question # 4 of 10 (Start time: 06:30:05 AM, 22 August 2021) Total Marks: 1

A _____ is a range of numbers inferred from the sample that has a certain probability of including e population parameter over the long run.

Select the correct option

- Probability limit
- Confidence interval
- Hypothesis
- Lower limit

STA301:Quiz 1

Question # 5 of 10 (Start time: 12:38:59 AM, 18 August 2021)

An estimator is said to be efficient if it has

Select the correct option

- ○ Smallest Variance
- ○ Unbiased estimate
- ○ Both (a) & (b) ✓
- ○ None of these

Question # 4 of 10 (Start time: 10:38:59 AM, 18 August 2021) **Total Marks: 1**

If we draw a sample of size 'n' from a population then sample is called "large sample" if:

Select the correct option

- ● n>40
- ○ n>30
- ○ n>50
- ○ n>25

In testing H0: $\mu = 100$ against H1: $\mu \leq 100$ at the 10% level of significance, H0 is rejected if:

- The p-value is greater than 0.10
- The p-value is less than 0.10
- 100 is contained in the 90% confidence interval
- The value of the test statistic is in the acceptance region

STA301:Quiz 1

Question # 10 of 10 (Start time: 12:44:16 AM, 18 August 2021)

_____ are equivalent

Select the correct option

- Type-I error and Level of confidence
- Type-II error and Level of confidence
- Type-I error and Level of significance
- Type-II error and Level of significance

$p = 0.16$, $q = 0.84$, $z = 1.96$

$e = 0.05$

$$n = \frac{(z_{\alpha/2})^2 \hat{p}\hat{q}}{e^2}$$

$$n = \frac{(1.96)^2 (0.16)(0.84)}{(0.05)^2}$$

$$n = \frac{0.5161}{(0.05)^2}$$

$$n = 206.52$$

$$n \simeq 207$$

Question # 1 of 10 (Start time: 04:39:31 PM, 22 August 2021)

If Var(T1) >Var(T2) ,where T1 and T2 are two unbiased estimators ,then:

Select the correct option

- T2 is more consistent
- T1 is more efficient
- T1 is more consistent
- T2 is more efficient

Question # 2 of 10 (Start time: 04:33:47 PM, 22 August 2021

The limitation of point estimation is that

Select the correct option

- ○ All of these
- ○ It is not calculated for non-normal distribution
- ○ There is no way of ascertaining how close the point estimate to
- ○ It is not calculated for small sample

STA301:Quiz 1

Question # 8 of 10 (Start time: 12:41:34 AM, 18 August 2021)

Alpha is the probability of .....

Select the correct option

- ○ Making Type II error
- ● Rejecting H1 when it is wrong
- ○ Making Type I error
- ○ Accepting Ho when it is true



Question # 5 of 10 (Start time: 06:31:12 AM, 22 August 2021) Total Marks: 1

Sample Proportion is a /an_____ estimator of population mean.

Select the correct option

- Unbiased and Consistent estimator
- Consistent
- None of these
- Unbiased

Question # 10 of 10 (Start time: 06:33:50 AM, 22 August 2021) Total Marks: 1

The width of the confidence interval is called_____ of the estimate.

Select the correct option

- Precision
- Unbiasedness
- Accuracy
- Biased

Question # 1 of 10 (Start time: 08:34:50 PM, 22 August 2021) Total Marks: 1

In interval estimation we obtained a ……of values as an estimate of parameter.

Select the correct option

- ○ Group
- ● Range
- ○ None of these
- ○ Both a and b

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 08:36:07 PM, 22 August 2021) Total Marks: 1

A confidence interval has a specified probability_____ of containing the true value of the parameter.

Select the correct option

- (●) 1-α
- ( ) 1+α
- ( ) α
- ( ) α/2

Question # 4 of 10 (Start time: 08:38:06 PM, 22 August 2021) Total Marks: 1

In testing H0: μ = 100 against H1: μ ≤ 100 at the 10% level of significance,H0 is rejected if:

Select the correct option

- The p-value is greater than 0.10
- The p-value is less than 0.10
- 100 is contained in the 90% confidence interval
- The value of the test statistic is in the acceptance region

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 08:39:41 PM, 22 August 2021) Total Marks: 1

"To consider every possible value that the parameter might have, and for each value, compute the probability and THAT value of the parameter for which the probability of a given sample is greatest, is chosen as an estimate." This procedure is known as

Select the correct option

- ○ The method of least square
- ○ The method of moments
- ○ The method of fractional moments
- ● The method of maximum likelihood

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 08:01:52 AM, 21 August 2021)

Total Marks: 1

In interval estimation we obtained a .....of values as an estimate of parameter.

Select the correct option

- Both a and b
- Range ✓
- Group
- None of these

Question # 9 of 10 (Start time: 10:28:12 PM, 20 August 2021)

Total Marks: 1

Conventionally, the probability of making a type-II error is denoted by:

Select the correct option

- ß (beta)
- ϴ (theta)
- a (alpha)
- б (sigma)

Question # 4 of 10 (Start time: 07:58:12 AM, 01 August 2021)

Total Marks: 1

A statistic whose standard error decreases with an increase in the sample size will be .....

Select the correct option

- Efficient
- None
- Unbiased
- Consistant ✓

Question # 10 of 10 (Start time: 08:08:31 AM, 21 August 2021) Total Marks: 1

If Var(T1) >Var(T2) ,where T1 and T2 are two unbiased estimators ,then:

Select the correct option

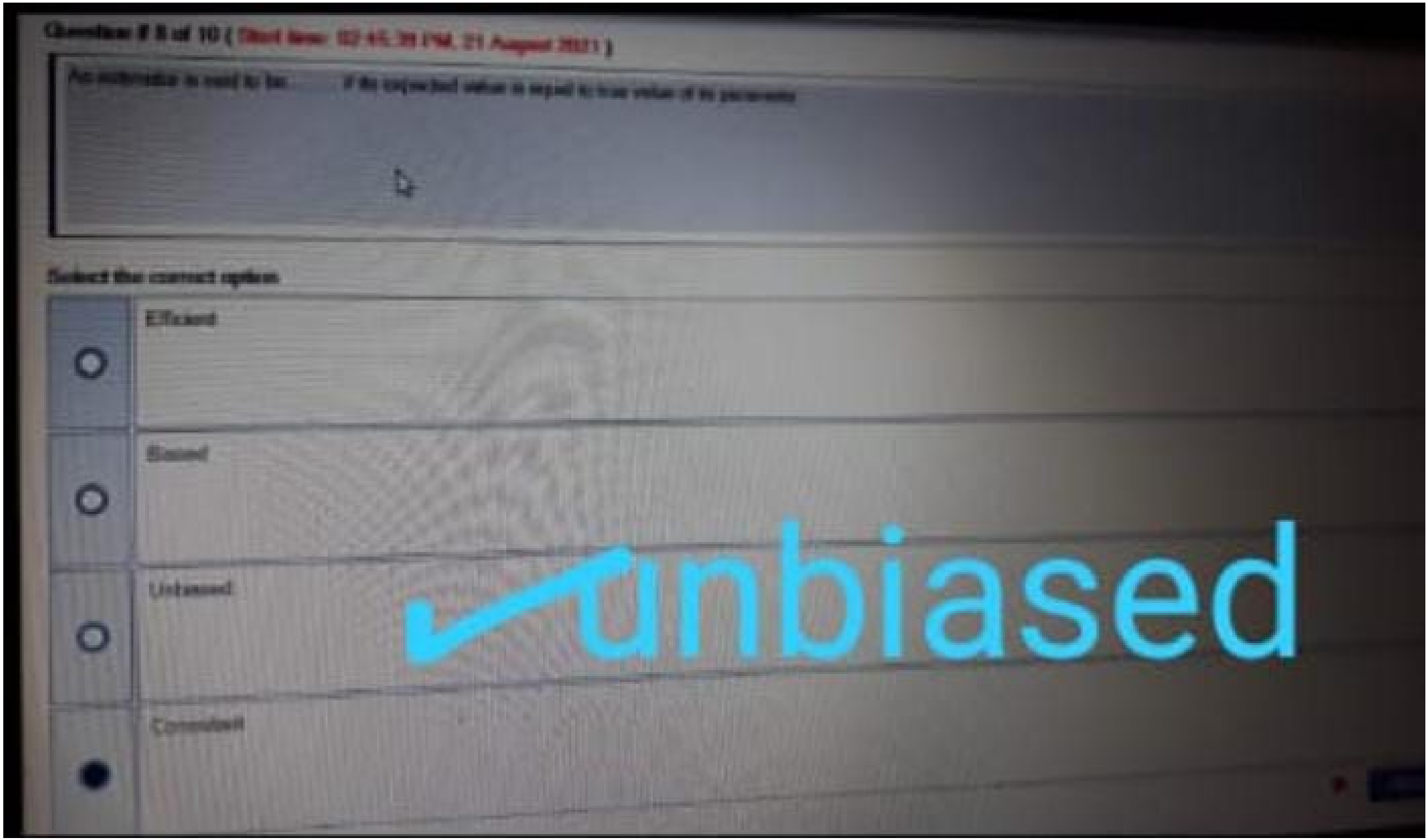
- T1 is more consistent
- T2 is more consistent
- T1 is more efficient
- T2 is more efficient ✓

Question # 7 of 10 (Start time: 02:19:09 PM, 20 August 2021)

Ideally, the width of confidence interval should be

Select the correct option

- ○ 0
- ● 1
- ○ 100
- ○ 98


unbiased

STA301Quiz 1

Quiz Start Time: 07:48 AM, 21 August 2021

Question # 3 of 10 (Start time: 07:58:01 AM, 21 August 2021)

Total Marks: 1

Sample mean is a/an ..... estimate of population mean.

Select the correct option

- Biased
- None of the above
- Unbiased ✓
- Either unbiased or biased

Question # 10 of 10 (Start time: 01:55:03 PM, 21 August 2021)

We can apply method of Maximum Likelihood on:

: correct option

- Qualitative variables only
- Discrete variables only
- Discrete as well as continuous variables ✓
- Continuous variables only

Suppose that the workers of factory B believe that the average income of the workers of factory A exceeds their average income, the hypothesis will be:

Select the correct option

- ○ MeanA > MeanB
- ○ MeanA ≥ MeanB
- ○ MeanA < MeanB
- ● MeanA ≤ MeanB

RIZ MUGHAL

Question # 7 of 10 (Start time: 08:02:37 AM, 21 August 2021) Total Marks: 1

Alpha is the probability of .....

Select the correct option

- Making Type I error
- Accepting Ho when it is true
- Rejecting HI when it is wrong
- Making Type II error

Question # 2 of 10 (Sets time: 07:10 AM, 01 August 2021)

Total Marks: 1

If we draw a sample of size 'n' from a population then sample is called "large sample" if:

Select the correct option

- n>30
- n>50
- n>40
- n>25

stion # 10 of 10 (Start time: 02:48:14 PM, 21 August 2021)

To determine a sample size in estimating population mean, we use the _____value

Select the correct option

- x
- z
- f
- t

Question # 5 of 10 (Start time: 01:50:29 PM, 21 August 2021) Total Marks:

An industrial designer wants to determine the average amount of time it takes an adult to assemble an "easy to assemble" toy. A sample of 16 times yielded an average time of 19.92 minutes, with a population standard deviation equal to 5.73. suming normality of assembly times, what is a 95%

e correct option

- (14.1123, 22.7277)
- (17.1123, 24.7277)
- (17.1123, 22.7277) ✓
- (17.1123, 30.7277)

Question # 5 of 10 (Start time: 07:57:49 AM, 21 August 2021)

Total Marks:

The method of Maximum Likelihood (ML) is used to find out:

No correct option

- Random numbers
- Sample values
- Point estimates
- Correlation coefficient

Question # 9 of 10 (Start time: 01:54:23 PM, 21 August 2021)

Confidence interval is also known as

Select the correct option

- Point estimator
- Level of Significance
- Margin of error
- None of these